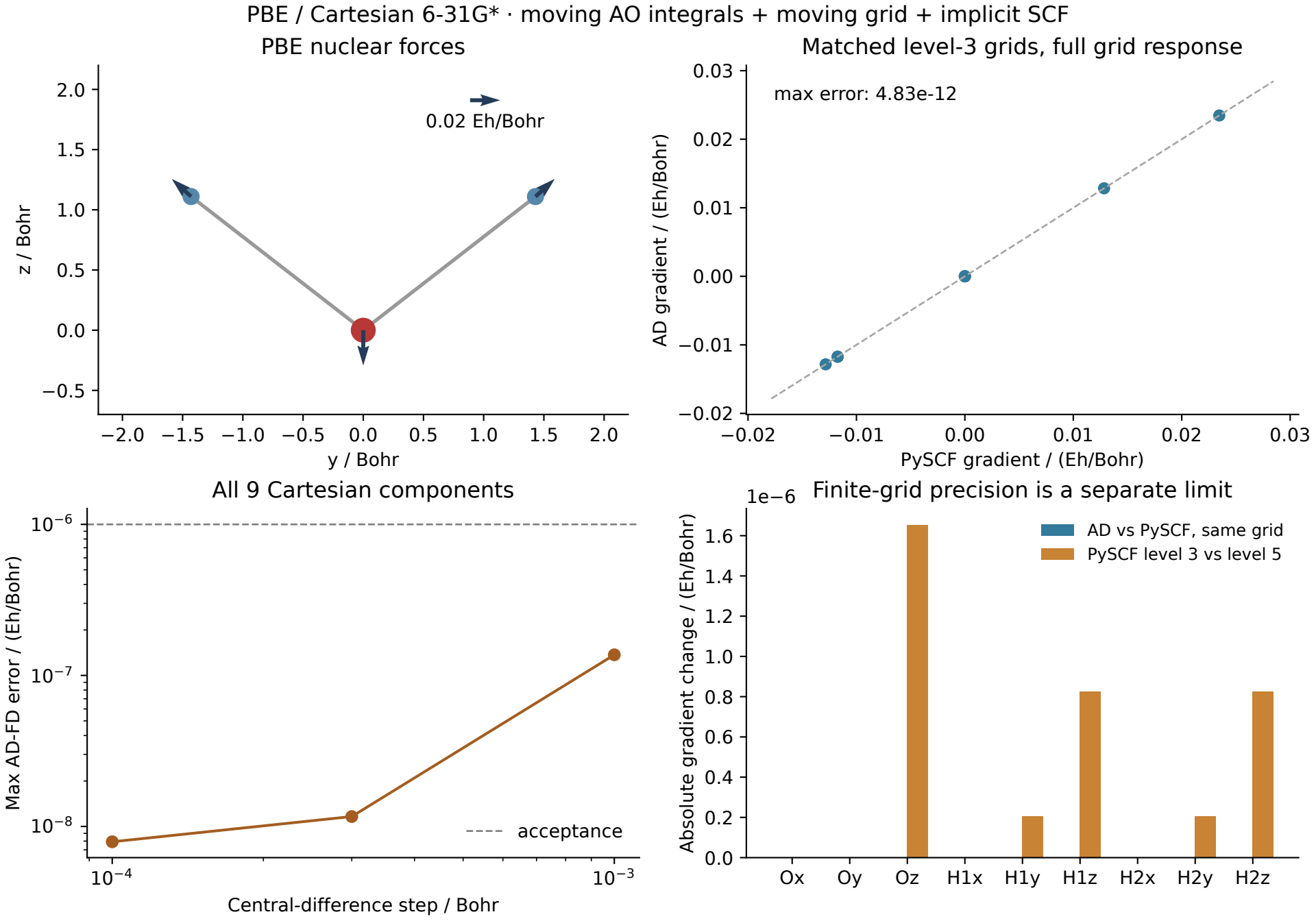
PBE / Cartesian 6-31G* · moving AO integrals + moving grid + implicit SCF
PBE nuclear forces
0.02 Eh/Bohr
z / Bohr
y / Bohr
Matched level-3 grids, full grid response
max error: 4.83e-12
AD gradient / (Eh/Bohr)
PySCF gradient / (Eh/Bohr)
All 9 Cartesian components
Max AD-FD error / (Eh/Bohr)
acceptance
Central-difference step / Bohr
Finite-grid precision is a separate limit
1e−6
Absolute gradient change / (Eh/Bohr)
AD vs PySCF, same grid
PySCF level 3 vs level 5
Ox
Oy
Oz
H1x
H1y
H1z
H2x
H2y
H2z